Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/23/2005
PI Name:	Pearlman, Howard G. Ph.D.		
Project Title:	The Cool Flames Space-Flight Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:	Howard Pearlman was formerly in the database as a PI at Drexel University. This is the same Howard Pearlman. He is now at GRC working with John McQuillen in PS.		
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-04
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC3-1006		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:

Solid fuel and premixed gas combustion exhibits some amazing self-organizaing patterns that rival those previously observed in the famous Belousov-Zhabotinskii liquid phase reaction and biological systems such as the electrical wave propagation observed in potential traces around the human heart. Such self-organizing patterns, including rotating spiral waves and target patterns, can produce instabilities and lead to nonuniform combustion and irregular pressure hstories. This can affect rocket motor thrust vs time profiles. Moreover, this study as much further reaching implications in that it may lead to an improved understanding of self-organizing structures that occur in a variety of chemical and physcal systems. Some have even suggested that such a mechanism may be responsible for LIFE on Earth and perhaps other planets! Understanding self-organizing structures is therefore imperative and will directly support our President's Exploration agenda.
In addition to understanding the mechanisms that lead to self-organization, cool flames are explored since they offer a new way to validate chemical kinetic mechanisms and an improve understanding is needed before newly developed engines (such as Homogeneous Charge Compression Ignition Engines) developed by most auto manufacturers become a reality. These engines offer the high efficiency of a diesel engine with the low emmissions common to spark-ignition engines. Imrpved control, especially at high loads, is important and needs to be addressed before such revolutionary concepts become a reality. To understand cool flames, the Cool Flames Experiment haad successfully passed its science concept review several years back and was scheduled to be developed for the ISS. In the past year, little progress has been made on the development side, while significant progress has been made in modeling these phenomena. The understanding of self-organization and cool flames are complimentary efforts with different objectives.

The first effort is directed at developing a greater understanding of the high-Lewis number diffusive-thermal instability in ground-based studies, exploring the role of Lewis number, gas mixtures, heat loss and dditives/inhibitors on system stability and reactive-diffusive structure. Experimentally, a large diameter tube filled with a mixture of known gases diluted with helium (one of the most prevalent the most prevalent elements belived to exist in the primordial gas that comprised Earth's atmosphere. The large diameter tube is important because the instability manifests itself as a long wavelength wave on the surface of an otherwise freely-propagating flame as the flame traverses the tube. To eliminate the complexities associated with the buoyant transport processes, reduced gravity testing is suggested to obtain an unobscured view of the competing diffusive processes of mass and heat.

The second effort aims to understand low-temperature oxidation reactions and cool flames. This is a flight-definition program. It considers a static reactor with prescribed mixture of known, premixed gases. The chamber and thus the gases are heated until a reaction, often a precursor to ignition, is initiated spontaneously, without spark or external ignition. In these systems, buoyancy alters the hydrodynamic, thermal and concentration fields. In turn, this alters the flame shape and reaction rate. Since the time required for reaction initiation can be on the order of minutes, or hours or even days(!), these experiments must be performed in long duration reduced gravity facilities. Such reactions have been the cause of extensive fire damage in art galleries, corn silos, grain piles, etc on Earth and it is critical to imporve our understanding of what conditions are safe and what conditions are not safe for short and long duration space flight. The amazing thing with cool flames is that the reactions can be extremely slow and take hours or days before the first sign of a fire is observed! During this time, however, ignition precursors are being formed, which can be detected in advance of the fire (autoignition event), thus preventing a fire in the first place. Prevention via early detection is no less important when it comes to fire safety then it is in medicine or other fields! Early detection saves lives! Instabilities similar to the diffusive-thermal instability to be studied have been observed in combustion synthesis of materials, as well as a host of other chemical and biological systems, including electrical propagation of signals in and around heart tissue (note, breakup of spatially organized waves is believed to be responsible for heart problems, most notably sudden cardiac death). In material synthesis, burning of a solid is involved, which mandates that the combustion can only be understood on the surface of the material. Since the current investigation involves gaseous components, transport throughout the system can be observed and parameters more easily varied. The knowledge thus gained, which is not available in any other configuration, should be directly applicable to combustion synthesis of materials as well as a providing a fundamental understanding of the couplings between reaction and diffusion, common to all combustion systems. The cool flame investigation is aimed at understanding processes that occur leading to spontaneous ignition. As such, the investigation will lead to an increased understanding of the physics behind knock in engines, which fuel manufacturers avoid by using additives, control of newly developed efficient engines and safety related concerns in spacecraft. This is lacking in current combustion theory.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Refined chemical kinetic models can be baselined using the data obtained from this experimental program. These refined models will be used to more accurately predict the combustion behavior of hydrocarbon fuels at low temperatures. It is relevant to autoignition in newly developed Homogeneous Charge Compression Ignition engines and also useful to mitigate potential fire hazards in space-based platforms. Prevention via early detection is no less important when it comes to fire safety then it is in medicine! Early detection saves lives!

Numerical / Computation Results
1. A five-step, skeletal chemical kinetic mechanism based on the original Gray-Yang model (the first model to predict the negative coefficient behavior of low temperature oxidation of hydrocarbons) has been developed. The model includes diffusive fluxes of heat and two intermediate species (a free radical propagating species and a branching species) and allows for initiation, branching, propagating and termination steps. It is a time-dependent, two-dimensional (spherical) model with axisymmetry assumed about the vertical diameter (no azimuthal variation in species concentrations, temperature of velocity). In addition, buoyancy convection is incorporated using the Buossinesq approximation in an otherwise assumed incompressible flow. This model is numerically solved and used to gain qualitative information on the behavior of cool flames at reduced-gravity, partial-gravity, and terrestrial conditions.

2. The model described in (#1) has been solved. Numerical results have been obtained on the two-dimensional temperature contours as a function of time for prescribed Arrhenius parameters (pre-exponential factors of reaction rates for each of the five steps, and appropriate activation energies) and assumed constant properties (cp) and diffusivities for heat and species concentrations. Species concentrations and velocity profiles remain to be analyzed. Ground-based (1g) studies have been performed using propane-air and propane-oxygen gaseous mixtures at select test conditions. The data has been analyzed and plots detailing the pressure history for different reactor temperatures, flame diameter versus time, and the associated ignition diagrams have been developed. Temperature contours predicted using modified Gray-Yang skeletal kinetic model at different instants in time; Temperature scale in K.

Task Progress:	3. A skeletal chemical kinetic mechanism developed by Wang and Mou to predict cool flame behavior in a CSTR (continuously stirred tank reactor) has been modified to include diffusive transport, a requirement for all unstirred, static reduced gravity studies. The model has been numerically coded and preliminary results have been obtained. For the first time, time-periodic, damped temperature oscillations are predicted at reduced-gravity. Experimentally, such temporal oscillations have been not (as yet) been observed at reduced-gravity for unity Lewis number within the available 20s on the KC135 aircraft, yet have been observed at high Lewis number (Le>>1) in butane-oxygen mixtures that have significant helium dilution. The oscillatory nature of cool flames at reduced-gravity is intricately linked with the conductive heat versus diffusive species transport. Additional experiments and numerical analysis is planned to further clarify the thermokinetic and diffusive couplings and explore the way these couplings lead to oscillations, sustained or damped. Cool Flames (Experimental Progress) 1.Cool flames in propane-oxygen and propane-air mixtures, based on the experiments conducted during the May 12th KC135 flight week at NASA GRC were very difficult, if not impossible, to image with our intensified cameras. Only one cool flame was imaged for a range of pressures, temperatures and fixed mixture composition (50vol%propane: 50vol.% oxygen). A more sensitive camera (CIR LLL camera) was subsequently tested and found to be sufficiently sensitive to image most equimolar propane: oxygen cool flames. 2. To determine the light emission spectrum associated with cool flames in propane-oxygen premixtures, an emission spectrometer was mounted such that the light emission, as viewed from the side window of the furnace (looking through the furnace windows and reactor vessel wall), was collected via a fiber optic cable and lens coupling In preliminary tests conducted using an equimolar propane:oxygen mixture at sub-atmospheric pressures and a fixed temperature (320oC), the spectrum could not be determined since the signal-to-noise ratio was inadequate (less than one). Averaging and increased integration times were tested, yet the spectrum remains undetermined. Based on this study, a high temperature fiber optic that can be inserted directly into the reactor is being pursued to eliminate transmission losses associated with the furnace windows and flask walls. 3. The CIR LLL-UV camera was shipped, setup and tested at Drexel University to determine it's ability to image low-light level cool flames in propane-oxygen premixtures at laboratory conditions. For nearly all conditions in the lab, the cool flames were successfully imaged, albeit in some cases with image-enhancing software. A new rack is being developed for the camera and support hardware in preparation for the next KC-135 cool flame experiment campaign. 4. Select experimental test points were conducted aboard NASA's KC135 microgravity aircraft (2 flight weeks) during microgravity, Martian and Lunar parabolic maneuvers using the same test mixtures. This data is currently being analyzed and directly compared to the 1g (laboratory) data to help clarify the role of the diffusive fluxes and weak buoyant convection (weaker than that on Earth) on cool flames and auto-ignition flame structure, emission, induction period and stability. 5. Ongoing collaboration with Professor Griffiths at University of Leeds to develop a comprehensive 1-dimensional cool flame model for propane: oxygen at low pressures has continued. This is in support of the current KC-135 aircraft studies and the future space-based experiments.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
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